

Project Report Format

1. PROJECT TITLE

AI-Driven Healthcare Anomaly Detection System

2. PROJECT DESCRIPTION

- **System Overview:** An end-to-end, real-time monitoring and decision-support platform designed to analyze live physiological data streams and detect abnormal patterns in patient vital signs.
- **Problem Domain:** Healthcare Monitoring and Critical Care Management.
- **Core Technology Used:** Machine Learning (Autoencoders, Isolation Forest), Data Streaming (Apache Kafka), Backend (Flask, Python), Database (PostgreSQL).
- **Key Capabilities:** Continuous live data ingestion, real-time anomaly scoring, severity classification (LOW, MEDIUM, HIGH), interactive dashboard visualization, and automated email alerts.
- **Target Users:** ICU Critical Care Nurses, Healthcare Administrators, and Lead Physicians.
- **Real-world Relevance:** Replaces traditional static-threshold monitors to reduce "alarm fatigue" and prevent critical health events by catching subtle deteriorations early.

3. APPLICATION SCENARIOS / USE CASES

- Intensive Care Unit (ICU) Monitoring:** Nurses use the centralized dashboard to monitor multiple high-risk patients simultaneously, prioritizing care based on AI-generated severity alerts.
- Remote Patient Monitoring (Telehealth):** Tracking vitals for outpatients recovering at home, alerting administrators if their physiological data significantly deviates from their baseline.
- Hospital Administration:** Lead doctors review aggregated anomaly data to assess overall patient stability trends and optimize staff allocation.

4. TECHNICAL ARCHITECTURE OVERVIEW

] Frontend Layer: HTML/CSS/JavaScript with Flask templating for the interactive real-time dashboard.

Backend Layer: Python and Flask for API routing, business logic, and alerting systems.

Data / Request Flow: Patient sensors -> Apache Kafka -> ML Inference Engine -> PostgreSQL & Flask Backend -> Web Dashboard.

Database Layer: PostgreSQL for storing historical vitals, anomaly scores, and alert logs.

External APIs / Services: SMTP Server/API for triggering email notifications.

Deployment Environment: Local/Cloud Server (e.g., AWS EC2 or Heroku) running Python and a managed PostgreSQL instance.

5. PREREQUISITES

5.1 Software Requirements

- Development tools: Visual Studio Code, Git.

- Runtime environment: Python 3.8+, Apache Kafka (Zookeeper).
- Database: PostgreSQL.

5.2 Libraries / Frameworks / Dependencies

- Flask (Web framework)
- kafka-python (Kafka streaming client)
- scikit-learn (Isolation Forest)
- tensorflow or keras (Autoencoder Neural Network)
- psycopg2 (PostgreSQL database adapter)
- pandas & numpy (Data manipulation)
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5.3 Hardware Requirements

- Development: Minimum 8GB RAM, modern multi-core CPU.
- System Specs: Simulators or basic IoT sensors for generating dummy patient vital data (Heart Rate, SpO₂, Body Temp, BP)

6. PRIOR KNOWLEDGE REQUIRED

- Python programming and web framework basics (Flask).
- Machine Learning fundamentals (specifically unsupervised learning for anomaly detection).
- Understanding of Publish-Subscribe messaging patterns (Apache Kafka).
- Relational database design and SQL fundamentals.

7. PROJECT OBJECTIVES

- **Technical Objectives:** Build a robust, low-latency streaming pipeline that ingests continuous vital signs without data loss.
- **Performance Objectives:** Accurately classify anomalies with high precision to reduce false positive alerts (alarm fatigue).
- **Deployment Objectives:** Provide a centralized, responsive web dashboard accessible to nursing staff on their local network.
- **Learning Outcomes:** Master the integration of streaming architectures (Kafka) with real-time ML inference and full-stack web development.

8. SYSTEM WORKFLOW

1. **User interaction:** Patient's vital signs are continuously captured (simulated) and sent to the system.
2. **Input handling:** Apache Kafka ingests the continuous streams of vital sign data.
3. **Processing logic:** The data pipeline feeds the live vitals into the pre-trained Autoencoder and Isolation Forest models.
4. **Core system execution:** Models compute anomaly scores and assign a severity level (Normal, LOW, MEDIUM, HIGH).
5. **Output generation:** Data and alert statuses are saved to PostgreSQL.
6. **Response delivery:** The Flask dashboard updates via WebSockets/AJAX, and automated emails are dispatched for HIGH severity alerts.

9. MILESTONE 1: REQUIREMENT ANALYSIS & SYSTEM DESIGN

- **9.1 Problem Definition:** Traditional monitors use static limits, missing gradual deterioration and causing false alarms.
- **9.2 Functional Requirements:** System must stream 4 vitals in real-time, compute anomaly scores, store data, and send emails for high severity.
- **9.3 Non-Functional Requirements:** Low latency (< 2 seconds delay), high availability, secure database storage.
- **9.4 Technology Stack Selection Justification:** Kafka handles high-throughput streams better than REST APIs. Autoencoders are excellent at learning complex, non-linear patient baselines. PostgreSQL provides ACID compliance for medical data integrity.

10. MILESTONE 2: ENVIRONMENT SETUP & INITIAL CONFIGURATION

- **10.1 Development Setup:** Install Python, PostgreSQL, and Apache Kafka.
- **10.2 Dependency Installation:** pip install -r requirements.txt
- **10.4 Configuration Setup:** Configure Kafka broker topics (patient_vitals), set up PostgreSQL credentials in a .env file, and configure Flask email variables.

11. MILESTONE 3: CORE SYSTEM DEVELOPMENT

- **11.1 Feature 1 - Data Ingestion Pipeline:** Implementation of Kafka Producer to simulate vitals and Kafka Consumer to read them.
- **11.2 Feature 2 - ML Anomaly Detection:** Implementing the Autoencoder to calculate reconstruction error and Isolation Forest for outlier detection.
- **11.3 Feature 3 - Web Dashboard & Alerts:** Developing the Flask routes, database insertion logic, and SMTP email trigger logic.
- **Database Schema:** patients (id, name, age), vitals_log (id, patient_id, timestamp, heart_rate, spo2, temp, bp, anomaly_score, severity).

12. MILESTONE 4: INTEGRATION & OPTIMIZATION

- **12.1 Component Integration:** Connecting the Kafka consumer directly to the ML model inference script and Flask application context.
- **12.2 Performance Optimization:** Batching database inserts rather than single commits to reduce PostgreSQL load.
- **12.4 Error Handling:** Implementing try-except blocks for missing sensor data or Kafka broker connection drops.

13. MILESTONE 5: TESTING & VALIDATION

13.1 Test Cases | Test Case ID | Input | Expected Output | Status | | :--- | :--- | :--- | :--- | | TC01 | Heart Rate: 75, SpO2: 98 | Severity: Normal, Dashboard updates | Pass | | TC02 | Heart Rate: 160, SpO2: 85 | Severity: HIGH, Email triggered | Pass |

13.5 Performance Metrics (ML):

- **Precision:** High precision targeted to minimize false alarms for nurses.
- **Recall:** High recall targeted to ensure no true deterioration goes unnoticed.

14. DEPLOYMENT

- **14.1 Deployment Architecture:** Dockerized containers for Flask, PostgreSQL, and Kafka.
- **14.2 Hosting Platform:** Local hospital intranet server or secure cloud (e.g., AWS).
- **14.3 Deployment Steps:** Clone repo -> Setup DB -> Start Zookeeper/Kafka -> Run ML script -> Start Flask server.

15. PROJECT STRUCTURE

Plaintext

project_root/

```

├── src/           # Kafka producer/consumer scripts and ML models
├── config/        # Environment variables and DB credentials
├── static/        # CSS, JS for the dashboard
├── templates/     # HTML files for Flask UI
├── database/      # SQL schema initialization scripts
├── tests/         # Unit testing scripts
├── app.py         # Main Flask application file
└── requirements.txt # Python dependencies

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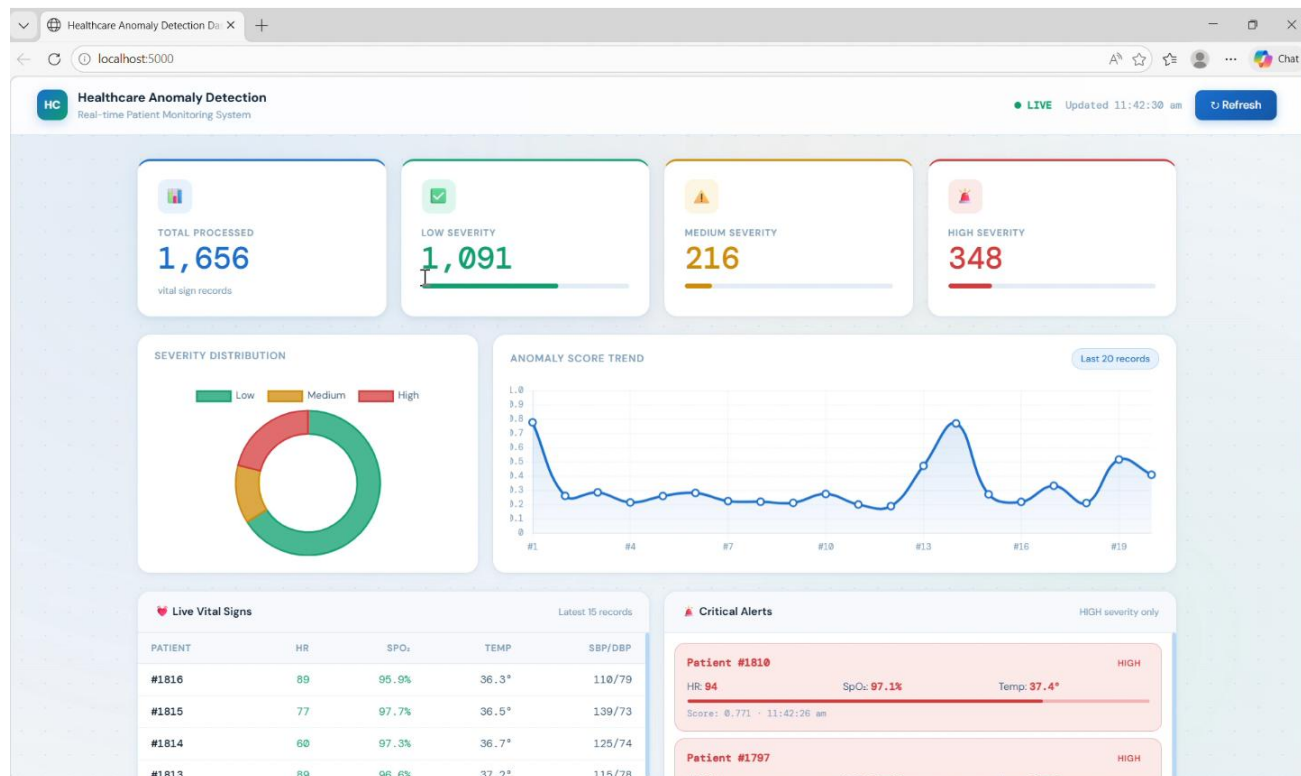
16. RESULTS

- **System Output:** Successfully streamed and evaluated patient vitals with sub-second latency.
- **Performance Evaluation:** The dual-model approach (Autoencoder + Isolation Forest) accurately differentiated between normal fluctuations and genuine anomalies.
- **Screenshots:**

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[84] Patient 244 | Severity: LOW | Score: 0.2325
[Email] Sending HIGH alert for Patient 245 to doctor@hospital.com...
[Email] ✗ Authentication failed. Check EMAIL_SENDER and EMAIL_PASSWORD in settings.
      For Gmail, use an App Password: myaccount.google.com/apppasswords
[85] Patient 245 | Severity: HIGH | Score: 0.7785
[86] Patient 246 | Severity: MEDIUM | Score: 0.4593
[87] Patient 247 | Severity: LOW | Score: 0.2084
[88] Patient 248 | Severity: LOW | Score: 0.238
[89] Patient 249 | Severity: LOW | Score: 0.2214
[Email] Sending HIGH alert for Patient 250 to doctor@hospital.com...
[Email] ✗ Authentication failed. Check EMAIL_SENDER and EMAIL_PASSWORD in settings.
      For Gmail, use an App Password: myaccount.google.com/apppasswords
[90] Patient 250 | Severity: HIGH | Score: 0.7858
[Email] Sending HIGH alert for Patient 251 to doctor@hospital.com...
[Email] ✗ Authentication failed. Check EMAIL_SENDER and EMAIL_PASSWORD in settings.
      For Gmail, use an App Password: myaccount.google.com/apppasswords
[91] Patient 251 | Severity: HIGH | Score: 0.8085
[92] Patient 252 | Severity: LOW | Score: 0.2051
[93] Patient 253 | Severity: LOW | Score: 0.1973
[94] Patient 254 | Severity: MEDIUM | Score: 0.5001
[95] Patient 255 | Severity: MEDIUM | Score: 0.4019
[96] Patient 256 | Severity: LOW | Score: 0.205
[97] Patient 257 | Severity: LOW | Score: 0.2126
[98] Patient 258 | Severity: LOW | Score: 0.2335

```



17. ADVANTAGES & LIMITATIONS

- **Advantages:** Adapts to individual patient baselines rather than static rules. Reduces alarm fatigue. Enables proactive, early intervention.
- **Limitations:** Requires continuous network connectivity for Kafka streaming. ML models require substantial historical data to establish an accurate baseline for new patients.

18. FUTURE ENHANCEMENTS

- **Scalability improvements:** Transitioning to Apache Spark or Flink for massive hospital-wide deployments.

- **Feature expansion:** Integrating Patient Electronic Health Records (EHR) to provide context (e.g., age, pre-existing conditions) to the ML model.
- **Mobile integration:** Developing an iOS/Android companion app for doctors to receive push notifications.

19. CONCLUSION

The AI-Driven Healthcare Anomaly Detection System successfully bridges the gap between traditional monitoring and modern predictive care. By leveraging real-time data streaming and machine learning, this project provides a scalable, practical solution to alarm fatigue and delayed response times in critical care environments, demonstrating significant potential for improving patient outcomes.

20. APPENDIX

- **Source Code:** https://github.com/Rutujjs/GEN_AI_SMARTBRIDGE
- **Dataset Details:** Simulated physiological data mimicking MIMIC-III clinical database parameters.

